Startup Studio · Columbia University · Fall 2026

Growth: channels and cold start

What to build next, and how to get people to it.

Class 10 · Friday, November 13, 2026 · 1:10 to 3:40 PM · Pupin 420

What we're doing today

Start

News and demos

AI news, the leaderboard, your ads, two teams present.

Build

What to build next

Rank what your tests found. AI in your product.

Grow

Channels

Where your next users come from.

Grow

Cold start

Network effects, seven strategies, what kills them.

Plan

Your plan

Your cold start plan, then homework.

The rest of the semester, 1 of 2

Date	Class	Topic	Homework due
Today	Class 10	Growth: channels and cold start	Usability tests, ad card, ad live
Nov 20	Class 11	Monetization and business models	Ad test results
Nov 27	No class	Thanksgiving	

The rest of the semester, 2 of 2

Date	Class	Topic	Homework due
Dec 4	Class 12	Startup 101 and pitch decks	Iterate, then freeze
Dec 11	Class 13	Pitch studio time	Pitch deck draft 1
Dec 18	Demo Day	Final pitches and live demos	Final deck

What you work on next, and why

Next class, Nov 20

- Class 11: Monetization and business models
- Ad test results
- Why it matters: How you will make money shapes what you build next. Unit economics shows whether the math works.

After that, Dec 4

- Class 12: Startup 101 and pitch decks
- Iterate, then freeze
- Why it matters: Investors expect you to know how startups get funded. Your deck turns what you learned into a story.

AI news this week

Jensen Huang resigned as Nvidia's CEO

- Thursday, Nov 12, effective immediately. He also left the board.
- CFO Colette Kress is interim CEO. A search for a permanent CEO is on.
- Nvidia gave no reason. Reports tie it to a board investigation.
- The stock fell about 4% that day. Earnings are Tuesday, Nov 17.

Nvidia 8-K, Nov 12 · CNBC

Microsoft's global outage, weeks after buyouts

- Thursday about 11 AM ET: Outlook down, Teams and Copilot disrupted.
- Problem reports from 74 countries. Not fixed 12+ hours later.
- Microsoft: not a cyberattack, not caused by the job cuts.
- On Oct 26 it offered buyouts to about 60,000 employees.

Al Jazeera, Nov 13 · AP, Oct 26

Where everyone is

Part 1

Your users, your ads, and two demos.

User count leaderboard

Go around the room. Update your numbers.

- Real users this week, next to last week's number.
- A real user signed up and did the core action once.
- 1 point per 10 real users. Full credit at 100.

User count leaderboard, 1 of 2

Where everyone is

Product	Last number you reported	Today
Aplovia (AIDI)	17 on waitlist	
CallIt (ZeroOne)	11 accounts, 9 confirmed	
Claude Daycare (Babysitters)	0 confirmed	
ClientReady (Stark Labs)	none yet	
Gable (Aditya)	none yet	

User count leaderboard, 2 of 2

Where everyone is

Product	Last number you reported	Today
AstraPath (Pola)	none yet	
Plandit (Black Panthers)	none yet	
Restroom Ready (Five Girls)	10 TestFlight downloads	
SettleWise (Jims)	0	
UnlockDrill (5 PPL)	none yet	

Ad check-in

One line per team.

- Which platform, and what question it answers.
- Is your experiment card committed?
- Is the campaign live? If not, what's blocking it?
- What's your pass bar?

Don't touch the ads until the 7 days end. The result is due next Friday.

What your weekly updates say

Your biggest blockers, in your words.

Blockers you named

- No real supply yet: listings are sample data.
- Growth came in below the forecast.
- Testers stall at setup. Only 2 of 6 finished unaided.
- A trust promise the new direction breaks.
- Users can't tell if their call was published.

Habits worth copying

- Saying why the week differed from the plan.
- Choosing testing over shipping, and saying so.
- Counting only what is real. "0 confirmed" is honest.
- Checking your own repo against your new plan.

Discuss

Which blocker is a cold start problem?

Look at your own update. Is your biggest blocker really about users?

Presentation schedule

10 minutes to present, then 5 minutes of questions.

Date	Class	Teams
Fri, Nov 13 (today)	Class 10	AstraPath, Settlewise
Fri, Nov 20	Class 11	AIDI, Black Panthers
Fri, Dec 4	Class 12	5 PPL, Aditya, ZeroOne
Fri, Dec 11	Class 13	Five Girls, The Babysitters

How the presentations work

Format

- Show the product live.
- Walk through your data.
- Where are you stuck?

You choose the focus

- Practice presenting a demo.
- Get feedback from peers on your product.
- You choose how to use your 10 minutes.

AstraPath

Presentation 1

Team Pola

Settlewise

Presentation 2

Team Jims

What to build next

Part 2

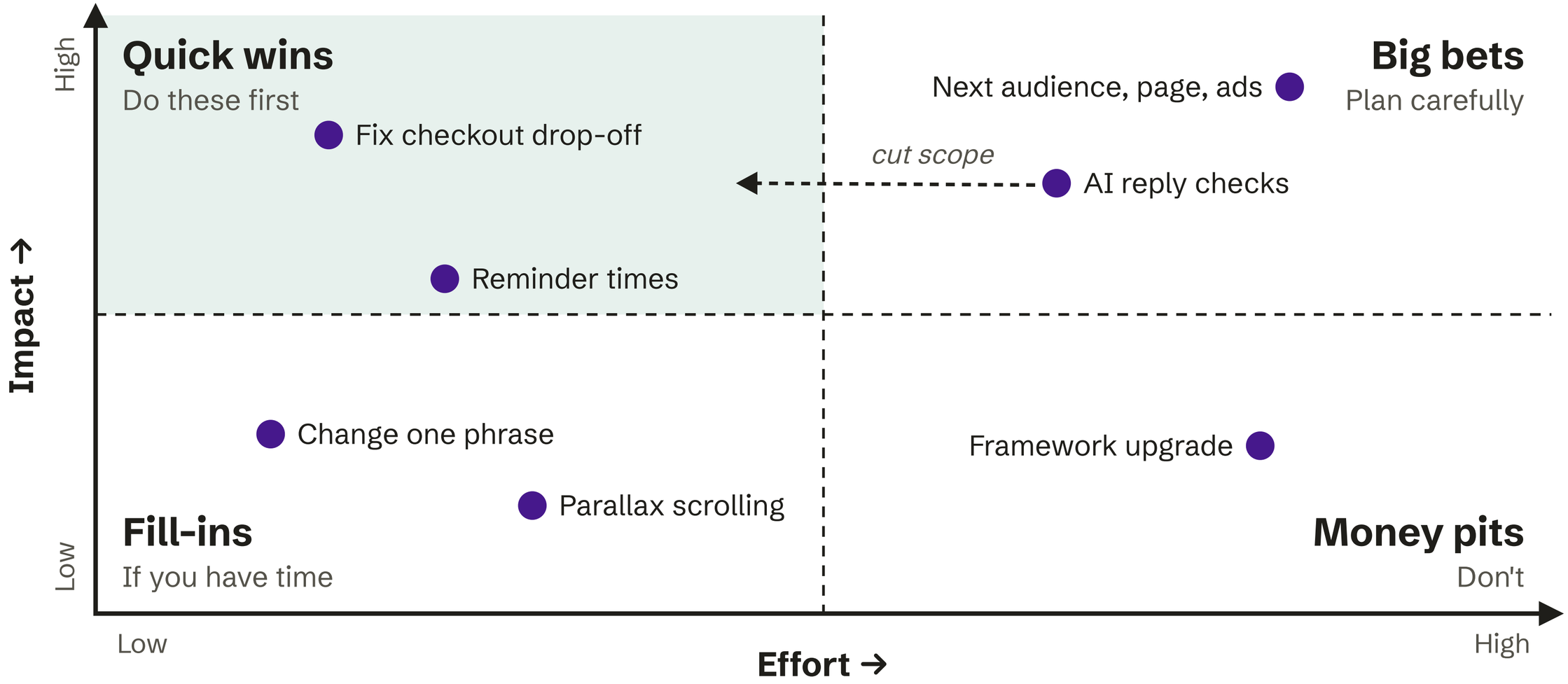
Your usability tests gave you a list. Here is how to decide what comes first.

Framework 1

Impact and effort matrix

Sort by gut. Fast and visual.

Impact and effort matrix



What does impact mean?

Define it first

Impact is whatever matters most to your goals. Revenue? User retention? Learning speed? Define it first, then tie every feature's impact back to those goals.

RICE scoring

Score by numbers. Makes your assumptions explicit.

RICE scoring

$$Score = \frac{Reach \times Impact \times Confidence}{Effort}$$

Created at Intercom. Compare relative scores. Don't overthink the numbers.

The four RICE factors

Factor	The question	Scale
Reach	Who and how much is affected?	1, 2, 3, 5, 8, 13
Impact	How deep is it for each person?	3 massive, 2 high, 1 low
Confidence	How sure are you of the other two?	100% data, 80% reasoning, 50% guess
Effort	Cost to build it safely	1, 2, 3, 5, 8, 13, 21

Big numbers are hard to estimate. The scale stops fake precision.

Rank your backlog on your own

Inputs

- Where your usability testers got stuck
- What you said you'd change in your synthesis
- Your own backlog

Outputs

- Top 5 to 7 items, ranked
- Your **#1 priority**, starred
- Bonus: tag items where AI could help

Knowing what you're not building is as important as knowing what you are.

Example: a startup's backlog

Task	Type	Priority
Fix checkout drop-off: add resume link	Bug	High
Let users set reminder times	Feature	High
Add quality checks for AI replies	Feature	High
Find the urgency angle for sign-ups	Growth	High
Pick next audience; build page, ads	Growth	High
Fix failing referral tests	Bug	Medium
Upgrade framework version for speed	Infra	Medium
Record first batch of short videos	Growth	Low
Change one phrase in AI replies	Quick win	Low
Add parallax scrolling to homepage	Quick win	Low

You can't do them all. What do you cut?

Real backlogs hold dozens of tasks. This one showed ten.

Part 3

AI in your product

Five patterns, and one rule.

LLMs aren't just chatbots.

Chatbots are one pattern. There are five or six others that are more useful for most of you.

Turn unstructured data into structured data

User writes a paragraph. AI extracts clean, structured fields.

Input

- "I'm looking for something fun to do Saturday night in the East Village, maybe live music, under $30"

Output

- neighborhood: East Village
- category: live music
- price_max: 30
- date: Saturday

Classify, sort, and route

AI looks at content and puts it in the right bucket. No conversation needed.

Events

Classify by vibe, genre, and audience.

User feedback

Route bug reports vs. feature requests vs. complaints.

Moderation

Flag inappropriate content before it's visible.

Let AI decide, let code execute

The LLM figures out what to do. Your code does it. This is called tool use or function calling.

```
# The AI hears: "User wants events in Brooklyn this weekend under $20"
# Your code runs:
search_events(location="Brooklyn", date="weekend", max_price=20)
```

Decision support

AI recommends. A human (or simple rules) decides.

Confidence	What happens
85% or more	Auto-execute
60 to 84%	Queue for review
Under 60%	Escalate to human

A stand-in for an algorithm you don't have data for

A trained model needs lots of data. A few examples and a clear prompt get you started today.

Recommend

Suggest events, places, or content that fit one person's taste.

Rank and score

Order apartments, leads, or applicants against stated priorities.

Match

Pair people with coaches, tutors, roommates, or each other.

Personalize

Write a plan, a summary, or a message around one person's goals.

Upgrade to a real ML model later, when you have the data.

Growth channels

Part 4

Where your next users come from, and how to reach them.

Your first 100 users come from people you reach by hand.

Paul Graham's "Do Things That Don't Scale" makes the case: early founders go out and recruit users themselves.

Six places first users come from

Channel	Best for	Watch out
Direct messages, in person	Your first 10 to 50 users	Slow. Each person needs real research.
Communities	People who already have the problem	Every group has its own rules. A ban ends the channel.
Campus	Products for students	Ask who may send to a list before you send.
Launch sites	A one-day spike of early adopters	Strict rules. Needs a working product.
Content and search	Trust over time	Slow. Most pages get no search traffic.
Newsletters, partners	Borrowed trust	You need an introduction. Few have public numbers.

Pick two or three

Score each channel 1 to 5 on four questions.

1. Can I name the place my user hangs out?
2. Can I reach them without breaking the rules?
3. Can I get a signal inside a week?
4. Do I have an honest reason to be there?

Keep one personal channel and one one-to-many channel.

Read the rules before you post

Where	What the rules say
Reddit	Each subreddit sets its own rules on self-promotion. Read the sidebar. Ask the moderators.
Show HN	No landing pages. Don't ask friends to upvote.
Product Hunt	Never ask for upvotes. Ask for feedback instead.
Discord	No unsolicited bulk messages. Each server adds its own rules.
LinkedIn	No promotional invitations to people you don't know.
Email	US law covers business email too. Say who you are. Honor opt-outs.

Every message has four parts

01	**Something specific about them**	A post they wrote, a question they asked.
02	**Who you are**	"I'm a Columbia student building X." Say it plainly.
03	**One small ask**	Try it for 3 minutes. Answer one question.
04	**An easy out**	"If it isn't useful, no worries."

You'd probably ignore this

Sent to 200 strangers at once.

Inbox noise

"Hey! We built an amazing app that revolutionizes how you discover events in NYC! Check it out at [link] and sign up for free! Let me know what you think!"

- ✕ Nothing specific to you
- ✕ Could be sent to anyone
- ✕ Asks you to sign up, with no reason why

You might actually read this

Same product. Sent to one person.

Feels personal

"I saw your post about weekend plans. I'm a Columbia student building a tool that picks NYC events by vibe. Would you try it and tell me if it sucks? Takes 2 min. No worries if not."

- ✓ Points at something they wrote
- ✓ Honest about who you are
- ✓ A small ask, and an easy out

Follow up once, then stop

- One follow-up after 3 to 5 days.
- One last note a week later.
- Answer "no" with thanks.
- Answer "stop" by stopping.

Most messages get no reply. Count yours and learn from it.

Start an inspiration folder

When a cold email, DM, or LinkedIn message actually gets you to pay attention, save it.

Screenshot it. Copy the text. Drop it in a folder.

When it's time to write your own outreach, open that folder. The best templates come from messages that actually worked on you.

Bonus: keep these as text, not just screenshots. Then you can feed the whole folder to an LLM and ask it to write outreach in that style.

"I'm a Columbia student building X"

People are often generous with students. Say who you are.

The plan: one card per channel

Your demand_strategy.md. Update it before class every Friday.

01	**Owner**	One name per channel. This week's person.
02	**Approach**	The community and its rules, the message, the tagged link.
03	**Volume**	How many touches this week. "10 DMs a day for 5 days," not "we'll post."
04	**Success bar**	Set before you start: touches, replies, signups, activated.

What happened: count four things

Per channel, every week.

01	**Touches**	Messages sent, posts made, flyers up.
02	**Replies or visits**	People who answered, or sessions from that channel's UTM.
03	**Signups**	From analytics, by UTM source and campaign.
04	**Activated**	Reached your aha moment. Signups alone are a vanity number.

Use AI to plan and to analyze

Plan

- "I'm building [product] for [audience]. What acquisition channels should I try first and how should I approach each?"

Analyze

- "Here are my results by channel. What worked, what didn't, and what should I focus on next?"

Five ways channels go wrong

Mistakes

- Posting the same link everywhere at once
- Counting likes and views, not signups
- One channel carrying the whole plan
- Never following up
- Never asking how people found you

Instead

- One community at a time, rules read first
- Count signups and activated users
- Two or three channels, each with a bar
- One follow-up, then stop
- One question in your signup form

Map your watering holes

Exercise, 12 minutes

12:00

1. Write where your users already gather online.
2. List the subreddits, Discord servers, newsletters, and groups.
3. Read each one's rules on self-promotion.
4. Score your top channels and keep 2 or 3.
5. **Hand in:** your picks and first message, in your demand strategy.

Network effects

Part 5

The theory you need before we get tactical.

What is a network effect?

Every new user makes the product more valuable for every existing user.

Network effect

- More users = more value for everyone
- Creates defensibility
- WhatsApp, Airbnb, LinkedIn

Not a network effect

- Virality = speed of user acquisition
- Scale = cost advantages from volume
- JibJab went mega-viral. Zero network effects.

~70%

of the value created by tech companies since 1994

NFX estimates this comes from network effects. They are the strongest form of competitive defensibility.

NFX estimate, from the spring cold start research report

Three laws of network value

Each law describes a different type of network.

Broadcast

$$V = N$$

Sarnoff's Law. One central node reaches many. Value grows linearly. Radio, TV, newsletters.

Communication

$$V = N^2$$

Metcalfe's Law. Every user can connect with every other. Phone networks, email, social media.

Group-forming

$$V = 2^N$$

Reed's Law. Groups can form freely. Value grows exponentially. WhatsApp groups, Slack.

Let users form groups, and value can grow exponentially. Not quadratic. But all three laws overestimate in practice.

Why all three laws are wrong

Useful as mental models, misleading as math. The real curve is closer to N · log(N).

01	**Unequal links**	Not all connections are equal. Your 1,000th Facebook friend adds almost no value.
02	**Human limits**	Humans can maintain about 150 meaningful relationships (Dunbar's number).
03	**Reed's Law**	At scale it is absurd. A network of 50,010 people cannot be 1,000x more valuable than 50,000.

What the laws miss entirely

01	**Phase 0**	All three laws start at N > 0. They say nothing about the most dangerous phase: getting from zero to critical mass.
02	**Engagement**	1M dormant users have less value than 100K active ones.
03	**User mix**	Real networks have buyers and sellers, creators and consumers.

Below a threshold, networks collapse. Above it, they self-sustain. A better model, from ecology. Beyond carrying capacity, growth decreases value.

Eflactem's Law

Metcalfe's Law in reverse. When users leave, value destruction is disproportionate.

Users	Value (N2)
200 users	40,000
100 users	10,000

50% user loss = 75% value loss

Why this matters

- Death spirals feel sudden but are mathematically predictable
- Users leave when staying costs more than it gives. Each departure pushes their friends toward the exit.
- Friendster's slide followed technical problems and a redesign that raised the cost of staying
- A 2013 study of Friendster's collapse mapped these cascades

Networks don't automatically improve with scale.

Past a certain point, growth actively destroys value.

Congestion and pollution

Congestion

- The network becomes overloaded.
- Highways at rush hour: more cars = slower for everyone
- Streaming during major events: servers overwhelmed
- Dating apps: most attractive users get flooded with messages and disengage

Pollution

- Low-quality participants degrade the experience.
- Social feeds flooded with spam and bots
- Marketplaces with too many low-quality listings
- Group chats past ~20 people: conversation quality collapses

Carrying capacity

The size where more users stop adding value to a network, and start taking it away.

What it looks like

- Past a point, each new user helps less, then hurts.
- Food delivery: conversion plateaus at ~45 restaurants in a 7-min radius
- Uber: once wait times hit 4-5 min, more drivers add negligible value

How to manage it

- Algorithmic filtering (News Feed over raw timeline)
- Trust systems (Airbnb reviews, eBay ratings)
- Access controls (privacy settings, workspace gating)

Strongest and weakest

Physical effects

- Telecom infrastructure
- Nearly impossible to displace

Bandwagon effects

- Hype-driven
- Can evaporate overnight

Most products layer multiple types.

Two types that matter for you

⇆ Direct network effects

- More users of the same type = more value.
- Every phone user benefits from more phone users
- WhatsApp, iMessage, LinkedIn connections

⇄ Indirect network effects

- More users of one type attracts a different type.
- More iOS users attracts more developers, which attracts more users
- More Uber drivers = better for riders (and vice versa)

Same-side negative effects are real. Growing one side can hurt that side.

Local vs. global network effects

Local (Uber)

- A driver in NYC only helps riders in NYC
- Every new city starts from zero
- Uber's massive US base gave them zero advantage against Didi in China

Global (Airbnb)

- A new host in Barcelona benefits travelers everywhere
- Competitors need global supply AND global demand simultaneously
- Result: Airbnb has few meaningful international competitors

If your network effects are local, your moat is local. Being #1 on one campus doesn't protect you on another. But if supply in one place benefits demand everywhere, scale compounds fast.

Network effects as a moat

NFX ranks the four ways a company becomes hard to kill.



1. Network effects

More users = more value. Self-reinforcing. ~70% of tech value since 1994.



2. Embedding

Switching costs. Integrations. Data lock-in. Oracle, Salesforce, Workday.



3. Scale

Cost advantages from volume. Economies of scale. Amazon, Walmart, AWS.



4. Brand

Psychological switching costs. Identity. Trust. Apple, Nike, Patagonia.

Most winning companies layer several. Network effects are the one that compounds on its own.

The cold start problem

Part 6

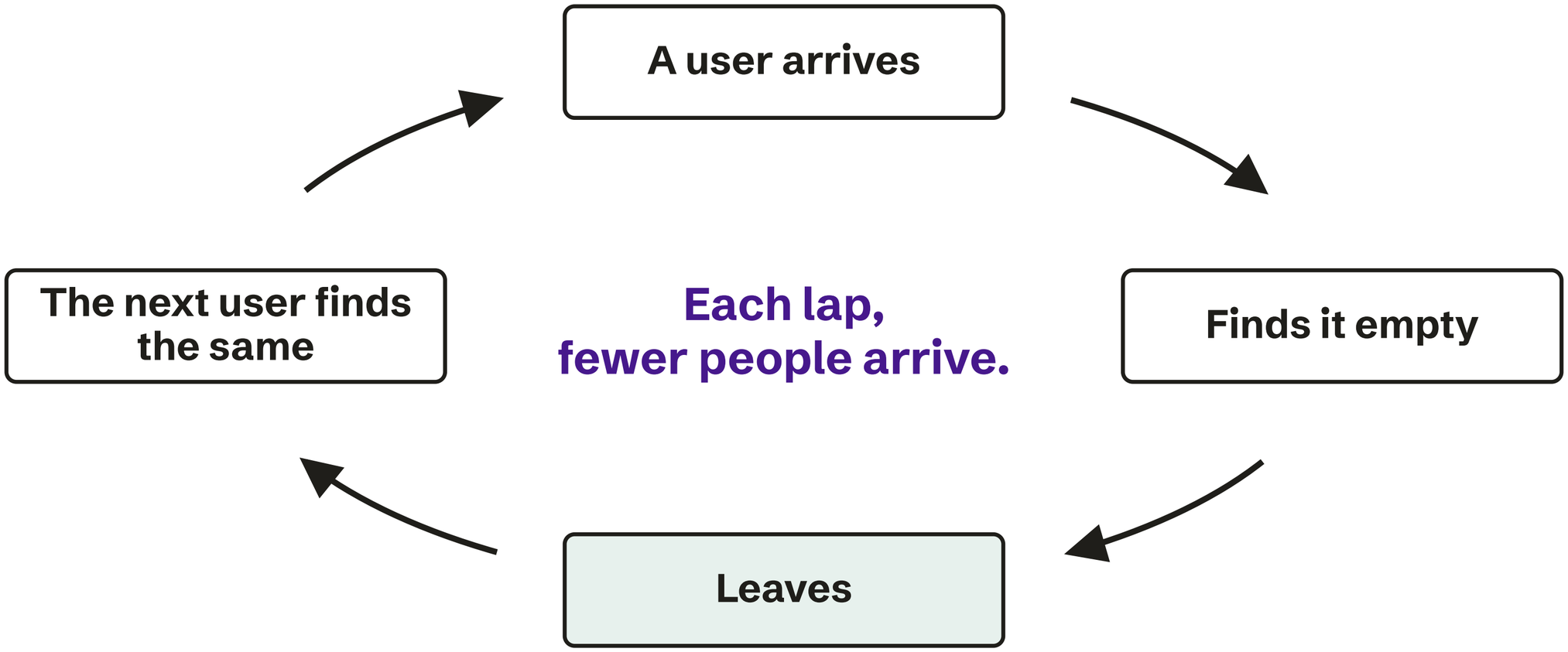
The hardest part: getting started when you have nothing.

The cold start problem

Your product needs users to be valuable.

But nobody wants to use a product with no users.

How an empty product dies



The default state

Default state	Before critical mass, the network **actively repels** users. This is the default state.
One shot	You only get **one shot** at a first impression. Users who bounce don't come back to check later.
The fix	The solution isn't "get more users." It's **get the right users, in the right density, in one place.**

Five types of cold start

Network

Two groups need each other. Dating apps, ride-sharing.

Content

Works, but empty. Reddit, Yelp, YouTube.

Marketplace

No buyers and no sellers. Airbnb, DoorDash, Etsy.

Group / invite

Useless until a whole group adopts. Slack, Zoom, games.

AI / algorithm

Recommendations need data you lack. Spotify, TikTok.

Quick diagnostic

The type decides the strategy. More than one yes means more than one type.

Ask	If yes, the type is
Two distinct user groups?	**Network** or **marketplace**
Empty without content?	**Content**
Needs a whole group at once?	**Group or invite**
Needs your users' data to work?	**AI or algorithm**

Your cold start types

Team	Type	Hard side
Gable (Aditya) NYC student sublets	Marketplace	Students listing sublets
AstraPath (Pola) Indoor routes to a room	Content + group	Buildings and departments
Restroom Ready (Five Girls) NYC restroom finder	Content / supply	Visitors who report details
Claude Daycare (Babysitters) Coding agents compete	Network + content	Owners who send agents
CallIt (ZeroOne) A friend group’s predictions	Group / invite	The group manager

Your cold start types

Team	Type	Hard side
Aplovia (AIDI) College search for international students	Content (light)	Trusted college data
UnlockDrill (5 PPL) An SAT question before you scroll	None	Takers who finish setup
SettleWise (Jims) Compare shortlisted apartments	Marketplace	Renters mid-search
Plandit (Black Panthers) Saved places into a day plan	None	Students with saved places
ClientReady (Stark Labs) Turns a client’s files into pages	None	Builders with a live client

Multiple cold start types? Solve the most painful one first.

Don't try to solve all of them at once.

The atomic network

The smallest group of users that can sustain itself without your constant intervention.

Product	Atomic network	Size
Zoom	2 people wanting to video call	2
Slack	1 company team	<10
Uber	Drivers and riders at one place and time (Caltrain station, 5 pm)	tiny

The atomic network

Product	Atomic network	Size
Facebook	Harvard's undergraduate student body	~6,800
Airbnb	~300 listings, ~100 of them reviewed, in one city	~300

Can you name your atomic network?

The smallest group that could use your product together, as real people.

Not yet

- "Anyone interested in X."
- You can't say who they are or where to find them.

An atomic network

- "The 15 people in my dorm who use a group chat to plan dinner."
- You could list their names and message them today.

Every network has a hard side.

The user group hardest to recruit, whose presence makes the product valuable for everyone else.

Hard side and easy side

Go after the hard side first. The easy side follows.

Product	Hard side (recruit first)	Easy side (follows)
Uber	Drivers	Riders
YouTube	Creators	Viewers
Tinder	Women	Men
Wikipedia	Editors (a tiny share)	Readers
Slack	Workspace admins	Team members

The 1/10/100 rule

Users

100

For every 100 users

Passive

10

engage passively

Creators

1

creates content. These are the hard side.

All early resources go to the 1%.

Density beats scale

- 50 users who all know each other are worth more than 500 who don't.
- A product that works perfectly for 50 people will spread. One that barely works for 5,000 will die.

Part 7

Seven strategies that work

Pick two. Execute with intensity.

The menu, 1 to 4

01	**Seed the supply**	Populate content yourself so it's not empty on arrival. You are the first 20 posts, listings, reviews.
02	**Constrained launch**	Go small to feel big. 100% of a small network beats 1% of a large one.
03	**Single-player mode**	Useful alone first. Come for the tool, stay for the network.
04	**Invite mechanics**	The product requires or rewards inviting others. Exclusivity creates demand.

The menu, 5 to 7

01	**Wizard of Oz**	Be the other side of the market yourself. Users see a working product; behind the scenes, it's you.
02	**Piggyback**	Attach to a platform where your users already are. Use their distribution.
03	**Event-driven launch**	Use a time-bounded event to concentrate density in one moment.

1. Seed the supply

Pre-populate the platform so users arrive to something real, not an empty room.

Who did it

- **Reddit.** Founders posted content from many fake accounts for months. Don't copy this today: it breaks Reddit's rules.
- **Yelp.** Created "Elite Squad," recruiting prolific local reviewers with exclusive events
- **GrubHub.** Manually scanned restaurant delivery menus and posted them before restaurants joined

For your team

- Write the first 20 reviews yourself
- Curate the first listings manually
- Scrape and import relevant content
- Don't launch social features until a new user sees immediate value

If a user opens your product and it feels empty, you haven't seeded enough.

2. Constrained launch

Launch to the smallest self-sustaining network. 100% of a small network beats 1% of a large one.

Who did it

- **Facebook.** Harvard only. About 4,300 accounts in two weeks. Then college by college.
- **Tinder.** Seeded at USC with about 300 people. Sorority chapters first, then the matching fraternity.
- **Slack.** One company team at a time. Teams that sent 2,000 messages had 93% retention.

For your team

- Pick one dorm, one club, one class, one company
- Get to density there before expanding
- Don't invite outside the group until it's clearly working inside it

Columbia gives you natural atomic networks: a cohort, a floor, a club. Use them.

3. Single-player mode

Build value for user #1 with zero other users. Add the network layer later.

Who did it

- **Instagram.** Photo filters were useful alone. 1M users in under 3 months.
- **OpenTable.** Sold restaurant management software, not the dining network. Booking layer came later.
- **Eventbrite.** Free self-service event pages. 17% converted to paid ticketing within 12 months.

Come for the tool. Stay for the network.

For your team

- Ask yourself:
- "What is my product worth to user #1, who knows no one else on it?"
- If the answer is "nothing," add standalone utility or change strategies

4. Invite mechanics

The product requires or rewards inviting others.
Exclusivity creates demand.

Who did it

- **Dropbox.** Both sides get free storage. Referrals were 35% of daily signups. 100K to 4M users in about 15 months.
- **Gmail.** Invite-only launch. Invites sold on eBay. Scarcity made people want in.
- **PayPal.** Cash bonuses for new accounts and referrals ($5 to $10 over time) + "Pay with PayPal" on eBay listings.

For your team

- You can't afford cash incentives. Lean on **exclusivity**: closed beta, hard cap of 30 spots.
- Build **collaboration-required invites**: features that only work when you invite someone.

5. Wizard of Oz

Manually be the supply side. Users see a working product; behind the scenes, it's you.

Who did it

- **DoorDash.** PDF menus + a phone number. Founders delivered a couple hundred orders themselves before Y Combinator.
- **Zappos.** Photographed shoes at retail stores, listed online, bought at retail when orders came in. Amazon bought it for about $1.1B.
- **Airbnb.** Rented a $5K camera, photographed apartments door-to-door. NYC revenue doubled in one month.

For your team

- If you're a marketplace: **do the supply side yourself**
- If it's content curation: curate manually
- If it's a service: perform the service
- Track what breaks. The manual process reveals what to automate next.

Strategy 5

Your users don't care if the supply is automated or manual.

They care if the product feels alive.

Some Wizard of Oz moments can be LLM calls

Pattern

If you're doing something manually behind the curtain, ask: could an LLM do this? Generating personalized recommendations? Summarizing user data? Drafting a weekly recap? A well-prompted API call might get you most of the way there.

6. Piggyback on existing networks

Attach to where your users already are. Read the rules first.

Who did it

- **Airbnb, Craigslist.** Reverse-engineered cross-posting. Listings looked more professional than native Craigslist and linked back to Airbnb.
- **YouTube, MySpace.** Built an embeddable video player for MySpace profiles. When MySpace tried to ban embeds, users revolted.
- **PayPal, eBay.** Pay with PayPal buttons on every listing. Every listing became a distribution channel.

Give value there before you pitch.

For your team

- Map your target users' existing platforms:
- Columbia Facebook groups
- Reddit r/Columbia
- Discord servers, WhatsApp groups
- LinkedIn

7. Event-driven launch

Use a time-bounded event to concentrate density in one moment.

Who did it

- **Airbnb, DNC.** 2008: 80,000 attendees, hotels booked solid. Targeted a moment of extreme housing demand.
- **Tinder, parties.** A party where the ticket was a Tinder download. Everyone in the room was on it.
- **Nextdoor.** Halloween: launched 3 weeks before, when neighbors naturally interact and need to coordinate.

For your team

- Pick a real event in the next 21 days:
- End-of-semester showcases
- Club events, hackathons
- Or create your own: a launch party with 30 people in a room

30 people gathered for your launch > 30 cold users discovered over 3 weeks.

What kills cold start solutions

Part 8

The most common mistakes. Some of you are making them.

Big bang launch

Releasing broadly before any dense community exists.

What happened

- **Google+.** Launched to hundreds of millions of potential users. By 2018, 90% of sessions lasted under 5 seconds. Never built one working atomic network.
- **Color.** Raised $41M and reached #2 in the App Store. The first screen was blank because no one nearby used it.

Why it fails

- Thin distribution everywhere, density nowhere
- Every first experience is an empty product
- They leave with a lasting negative opinion

Instead: a constrained launch. Go small. Go dense.

Not focusing locales

Optimizing for total users instead of connections between users.

What happened

- **Sidecar.** Spread across about a dozen cities. Critical mass in none. First to ride-sharing, and lost.
- **Uber.** Saturated one city at a time. Won city after city, though not everywhere (it sold China to Didi).

Why it fails

- Users in a sparse network never meet each other
- The product feels empty even with a large count
- No single market reaches the tipping point

Network value comes from connections, not users. 1,000,000 users with 2 connections each is worth less than 100,000 with 30.

Wrong-side seeding

Starting with demand before you have supply.

What happened

- **The research.** 14 of 17 successful marketplaces in Lenny Rachitsky's research focused on supply first.
- **Supply pulls demand.** Creators share their pages. Hosts list on several platforms.

Why it fails

- You drive traffic to a site with nothing on it
- Users bounce and form a negative impression
- Supply never sees demand, so it doesn't join either

Build supply first. Let demand find a product with something to offer.

Discounts to solve cold start

Homejoy raised $40M and offered $19 cleanings against $85 at market rate. It's dead.

What happened

- **75%** of bookings came from discounts (reported).
- **25%** month-1 retention, and 10% or less at six months.

Why it fails

- Discount users came for the deal, not the product
- They leave when the price normalizes
- You measure price sensitivity, not product-market fit

You can't discount your way out of a density problem.

Hard side neglect

Not giving enough value to the users who make the product work for everyone else.

What happened

- **Clubhouse.** Exploded in lockdown, with celebrity speakers. No way to pay speakers until April 2021. Its best speakers left within months.
- Daily active users fell 68% from the Feb 2021 peak by May 2021 (Apptopia).

Why it fails

- The hard side creates the value everyone else consumes
- If they aren't rewarded (money, tools, status), they leave
- Then the easy side has no reason to stay

Check yours: who does the work that makes your product valuable, and what do you give them?

What I'm seeing in your work

Good instincts

- You counted who reached the aha moment, tester by tester.
- You wrote what testers did. Searched for room 502, got a water fountain.
- You fixed the step right before the aha moment. Setup went from 8 steps to 3.
- You wrote down what you're not changing, and why.
- You tested with strangers, not just friends.

Watch out for

- Only if answers. Only if my building is covered. Only if my friends use it. That's your cold start.
- Sign in before value. Testers clicked Sign in before they had an account.
- Labels only your team can read. "3 of 3" read as three bedrooms.
- No experiment card or live ad in a pull request yet. The card goes in before you spend.

Match your strategy to your problem

Seven strategies

Your situation	Primary	Secondary
Empty platform, no content	Content seeding	Single-player mode
Marketplace, no supply	Wizard of Oz	Constrained launch
Network product, nobody to connect with	Constrained launch	Invite mechanics
Good product, no distribution	Piggyback	Event-driven launch
Need a whole group to adopt	Invite mechanics	Event-driven launch

Supply first, almost always.

Your cold start plan

Part 9

Time to apply this. Four questions, then a plan you can start this week.

Answer four questions

01	**Cold start type**	What is your cold start type? Network, Content/Supply, Marketplace, Group/Invite, AI/Algorithm, or none?
02	**Hard side**	Who is your hard side? Name specific people, not demographics.
03	**Two strategies**	What are your two strategies? Use the decision table. Pick a primary and a secondary.
04	**This week**	What can you do THIS WEEK? One concrete action for each strategy. Not "think about it." Do it.

Your cold start plan

Exercise, 10 minutes

10:00

1. Name your cold start type.
2. Name your hard side: specific people, not demographics.
3. Pick two strategies from the decision table.
4. Write one action for each strategy that you can do this week.
5. Write it in your team doc. You will finish it for homework.

What's due and when

Due	What	Details
Today, Nov 13	Homeworks 7 and 8	Usability tests; ad card and campaign live
Fri, Nov 20	Homework 9	Ad result, cold start plan, backlog, demand strategy
Every Friday	Weekly update	Before class, in your team folder

The only test that matters

Try it this week

Give your product to someone who doesn't know you and has no obligation to be nice. Watch what they do. Do they come back the next day?

Homework 9

Your ad result, your cold start plan, and what you build next.

What to submit by Friday, November 20

Due at 1:10 PM. One pull request per team to your team folder.

1. **Part 1:** your ad result
2. **Part 2:** your cold start plan
3. **Part 3:** your backlog, ranked
4. **Part 4:** your demand strategy, updated

Then paste the pull request link into Courseworks.

Your ad result

Section 9 of your experiment card, after your 7 days end. Numbers, then what they mean, then what you'll do.

01	**Numbers**	Impressions, clicks, signups, spend. Google Analytics by UTM, with a screenshot.
02	**Pass or fail**	Against the bar you wrote before launch. Not a new bar.
03	**What's next**	What you'll do because of it.

Even if your numbers are small

5 users is enough to learn from.

- 5 signups and 0 completed onboarding? Clear signal.
- 3 of 5 users from the same channel? Clear signal.
- Nobody clicks the button you thought was obvious? Clear signal.

Don't wait for statistical significance. Look for patterns and act on them.

Your cold start plan

Homework 9, part 2

01	**Cold start type**	Network, Content/Supply, Marketplace, Group/Invite, AI/Algorithm, or none.
02	**Hard side**	Name specific people, not demographics.
03	**Two strategies**	A primary and a secondary, from the decision table.
04	**This week**	One concrete action for each strategy.

Your backlog and your demand strategy

Homework 9, parts 3 and 4

01	**Your backlog**	backlog.md: today's ranking, your #1 starred, and what you're not building.
02	**Demand strategy**	Updated with this week's channels and your first messages.

Questions?

Density beats scale

Pick two strategies. Execute this week.

Next class · Friday, November 20, 2026 · 1:10 to 3:40 PM · Pupin 420